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Types of Ensemble Learning:

4 main important types:
a) Voting ensemble
b) Bagging → [Random Forest]
c) Boosting → [Adaboosting, Gradient boosting, XgBoost]
d) Stacking

a) Voting Ensemble.

- Base models are different algorithm, getting same data.

testing
- regression: take mean value
- classification: majority wins.

assumption:
- ~~combined mean~~ all model of accuracy of all model should be > 51%
- all model should be independent.

otherwise worst answer comes.

Let 3 ~~mo~~ different independent models m_1, m_2, & m_3.

~~accuracy | prop of g~~

~~Accuracy score~~

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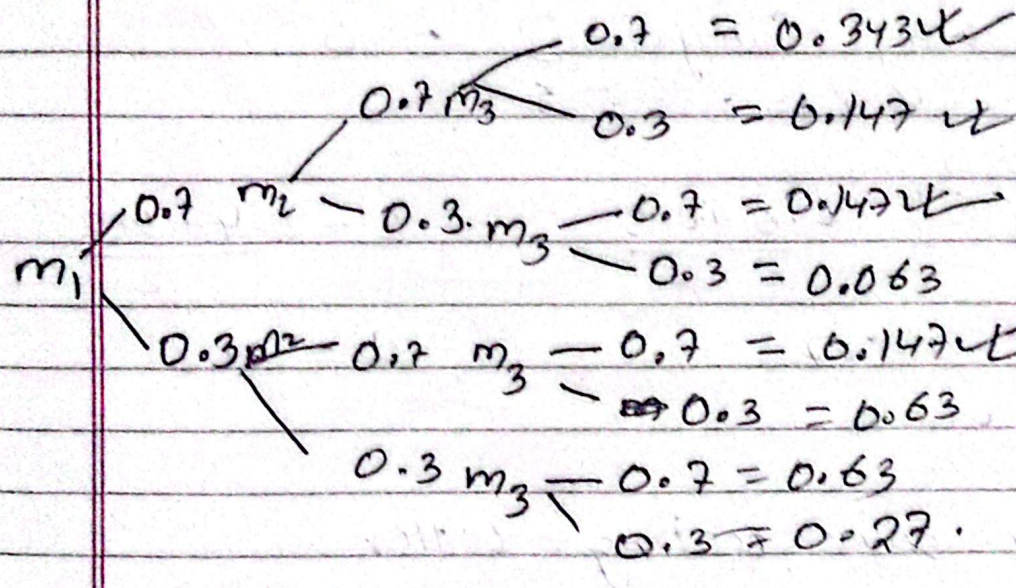
probability of getting correct | probability of getting wrong answer.

= accuracy score | 1 - accuracy score.

Base model

m_1	m_2	m_3
0.7 | 0.3	0.7 | 0.3	0.7 | 0.3



getting ~~correct~~ correct by majority.
(2 → 0.7 or more)

0.343
0.147
0.147
0.147
0.78

≈ 78% correct accuracy.

- If all will be less than 51 then the condition will be worst.

~~Hard voting~~

$m_1 \to 0 - 0.6$
$\quad\quad\;\; \to 1 - 0.4$

$m_2 \to 0 - 0.8$
$\quad\quad\;\; \to 1 - 0.2$

- Hard voting [majority] $\to 0$
  ~~0.6 , 0.8~~
- Soft voting
  $0 - \frac{0.6+0.8}{2} \quad , \quad 1 - \frac{0.4+0.2}{2}$
  $= 0.7 \quad , \quad 1 - 0.3$

- Generally soft voting better.
  But test one.

code to check accuracy of each model before using the Voting ensemble.

```
estimator = [('lr', clf1), ('rf', clf2), ('knn', clf3)].

for estimator in estimators:
    n = cross_val_score(estimator[1], X, y, cv=10,
                        scoring='accuracy')
    print(estimator[0], np.round(np.mean(n), 2))
```

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```
from sklearn.ensemble import VotingClassifier
vc = VotingClassifier(estimators= estimators, voting='soft')
Print( np.mean( cross_val_score( vc, X, y,
        cv=10, scoring="accuracy").
```

voting = 'hard' default.

```
i, j in range(1,4):
vc = VotingClassifier(estimators = estimators,
        voting = 'soft', weight=[i, j, k]
```

weight = importance changes.

Voting Ensemble Regression

taking mean

```
from sklearn.ensemble import VotingRegressor
```